

ArunKumar Kandasamy

Principal Software Engineer

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Forward-focused **Principal Software Engineer** with 16+ years of experience designing and delivering scalable web platforms and enterprise products across Agile and Hybrid delivery models. Blends hands-on architecture and engineering leadership (Ruby on Rails, cloud, performance, distributed systems) with end-to-end delivery ownership (planning, execution, risk management, and release governance) to drive predictable outcomes. Experienced in partnering with Product, Engineering, QA, Support, and Marketing across the SDLC, including beta programs and production releases.

Distributed Systems | Scalability | High Availability | Microservices | API Design | Cloud-Native Architecture

TECHNICAL SKILLS

Language: Ruby,
JavaScript

Backend & APIs: Ruby on
Rails (v3-v7+), REST APIs,
GraphQL

Frontend: React, Next.js,
Angular, Tailwind CSS

Cloud & DevOps: AWS,
Docker, CI/CD (Jenkins,
GitHub Actions, CircleCI),
Capistrano

Databases & Caching:
PostgreSQL, MySQL,
MongoDB, Redis

**Messaging & Async
Processing:** Sidekiq,
Delayed Job, ActionCable,
Turbo Streams, ActiveMQ

**Architecture &
Integrations:** Distributed
Systems, API Integrations,
Event-Driven Architecture

Testing & Quality: RSpec,
Jest, React Testing Library

Monitoring & Reporting:
NewRelic, JSReport, D3.js

AI & Automation Tools:
Claude, Cursor IDE, LLM-
based coding Assistants.

PROFESSIONAL SUMMARY

- **Strategic Architecture:** Translating complex product roadmaps into scalable, high-performance Ruby on Rails ecosystems that accelerate time-to-market.
- **Operational Excellence:** Improving reliability and uptime through expert-level database tuning, observability (APM), and proactive system hardening.
- **Technical Stewardship:** Bridging the gap between stakeholders and engineering through crisp decision records (ADRs) and Agile delivery ownership.
- **Scale-First Mentality:** Designing API-first, distributed systems that balance immediate feature velocity with long-term architectural integrity.

CORE SKILLS / EXPERTISE

- **System Design:** Leading HLD/LLD, system reviews, and trade-off analysis for enterprise-grade platforms.
- **Performance Engineering:** Reducing p95 latency and optimizing database costs through advanced caching and query profiling.
- **Scalable Backend:** Designing event-driven workflows (Sidekiq/Kafka) and complex domain models (PostgreSQL).
- **Modernization:** Leading zero-downtime Rails/Postgres upgrades and monolithic-to-service refactoring.
- **Delivery Leadership:** Owning the full SDLC: from discovery and risk management to CI/CD and production readiness.
- **API Ecosystems:** Building secure, versioned REST/GraphQL contracts with RBAC/auth patterns.

AI-AUGMENTED ENGINEERING PRACTICES

- Follow AI-assisted software engineering practices across application development and DevOps workflows by leveraging Claude, Cursor IDE, and LLM-based coding assistants for architecture design, code generation, refactoring, test automation, and technical documentation.
- Apply AI capabilities within Dockerized environments and GitHub Actions to streamline CI/CD pipelines, automate code quality checks, security scanning, release validations, and deployment processes, with a focus on improving development velocity, deployment reliability, and operational efficiency.
- Follow prompt engineering and AI code review best practices to maintain consistency, security, and quality standards.

EXPERIENCE HISTORY

Principal Software Engineer

ThinkPro Technologies, Chennai | Jan 2024 - Present

Lead engineering direction, owning architecture (HLD/LLD), execution planning, and end-to-end delivery across internal products and client engagements. Consulted for enterprise clients (**Cornerstone OnDemand, Effortless FinLabs**), leading legacy modernization initiatives and delivering measurable performance improvements. Architected and launched VoiceHunt.news using a decoupled Rails API + Next.js architecture, enabling scalable, SEO-optimized content delivery and API-driven content lifecycle management. Improved p95 latency by 41% through caching, revalidation, and database optimization, while establishing CI/CD pipelines (Docker, Jenkins, GitHub Actions) for reliable deployments. Drove delivery execution, production stability, and release governance across the SDLC, including planning, risk management, and post-release validation. Mentored engineers and enforced architecture standards and code quality.

Technical Lead

CAI Info India Private Limited, Bangalore | Feb 2020 - Mar 2023

Led backend engineering and technical execution for enterprise platforms using Ruby on Rails, collaborating with cross-functional stakeholders to deliver scalable, business-critical systems. Owned end-to-end SDLC including architecture discussions, delivery planning, release readiness, and production support, while building REST APIs and asynchronous workflows (Sidekiq/Delayed Job) for reliable processing. Improved system performance by reducing p95 latency by 28% and DB load by 22% through query optimization and refactoring, and strengthened release stability via CI/CD (Jenkins, CircleCI), Docker-based deployments, and automation. Enhanced system reliability and security using RBAC, environment configuration, and production issue handling, reducing **MTTR by 38%**, while driving code quality through reviews, debugging, and design leadership.

Senior Software Engineer

ZiniosEdge Information Technologies Private Limited, Bangalore | April 2018 - Jan 2020

Developed and scaled a large CRM platform using Ruby on Rails, delivering modules across customer lifecycle, workflow automation, and reporting. Built and maintained REST APIs with clean service boundaries, and integrated analytics workflows (MicroStrategy, JSReport) for automated insights. Improved database performance by ~30% through schema design, indexing, and query optimization across PostgreSQL/MySQL. Contributed across the full stack (JavaScript, HTML, CSS/Sass) and owned end-to-end feature delivery, consistently releasing 12+ features per quarter across multiple stakeholders.

Senior Software Engineer

Aspire Software Consultancy, Bangalore. | Nov 2016 - Jan 2018

Collaborated with a UK-based B2B marketplace (Rightboat) as a senior engineer, owning end-to-end delivery of product enhancements using Ruby on Rails. Built backend workflows, optimized database interactions, and improved system reliability through performance tuning and bug fixes. Partnered directly with stakeholders on requirements, solution design, and delivery timelines, while managing deployments and production support to ensure stable, high-quality releases.

Co-Founder & Engineering Lead @ Code Rabbits Technologies | Senior Software Developer @ Mereo Technologies → Software Developer @ Aubev Technologies → SEO Analyst @ Spoison Consulting, Chennai. | Oct 2009 - Oct 2016

Founded and led engineering for a SaaS-based eCommerce platform, designing modular architecture using Ruby on Rails and owning end-to-end delivery across web and mobile solutions. Progressed from SEO Analyst to Software Developer and Senior Developer, building a strong foundation in full-stack development, system design, and production support. Delivered multiple software projects, improved system stability through debugging and performance tuning, and collaborated with cross-functional stakeholders to build scalable, maintainable systems.

KEY PROJECTS

GOEFFORTLESS.AI | April 2024 - March 2026 — Hybrid Accounting & Distributed Data Sync Platform

Role: **Lead Software Consultant** | Stack: **Ruby on Rails, React, PostgreSQL, GCP, gRPC, Windows Service (.NET), Kafka, Sidekiq, REST APIs**

- Designed hybrid sync architecture bridging on-prem Tally DB with cloud services using a Windows-based ETL bridge.
- Implemented data pipelines using batch processing, delta sync, checksum validation, and idempotent writes.
- Built ingestion APIs using Rails + gRPC adapters, enabling high-throughput financial data processing.
- Implemented event-driven workflows using Kafka for downstream processing and audit trails.
- Designed async processing using Sidekiq + Redis, with retry policies and deduplication logic.
- Implemented reconciliation workflows for eventual consistency and financial data integrity.

VOICEHUNT.NEWS | Jan 2025 - Present — Distributed Content & Edge-Cached Publishing Platform

Role: **Engineering Lead** | Stack: **Ruby on Rails, Next.js (SSR, ISR), PostgreSQL, Redis, GraphQL, AWS (CloudFront, S3, SQS, DynamoDB, Lambda@Edge), Docker, Nginx, RSpec, Jest, React Testing Library, New Relic**

- Designed decoupled architecture using Rails API + Next.js SSR/ISR, enabling hybrid rendering and SEO-optimized content delivery.
- Implemented Edge caching + ISR revalidation using CloudFront + Lambda@Edge, with cache state persisted in DynamoDB and invalidation triggered via SQS + Lambda.
- Built API layer with GraphQL + REST, optimizing payload size and reducing over-fetching.
- Implemented backend performance optimizations using ActiveRecord scopes, query planner analysis, Redis-based fragment caching, and materialized counters.
- Containerized services using Docker + Nginx reverse proxy, deployed on EC2 + RDS, with static assets on S3.
- Implemented test strategy using RSpec (API + service layers) and Jest + React Testing Library (SSR/ISR hydration + UI regression).
- Observability via New Relic APM, custom metrics, slow query tracing, and memory profiling; improved cache hit ratio to 91% and reduced TTFB by 52% on high-traffic pages.

EDCAST.COM | Jan 2024 - Nov 2025 - *Enterprise Platform Modernization & Rails Upgrade Program*

Role: **Lead Consultant** | Stack: **Ruby on Rails, Node.js, PostgreSQL, React, RSpec, FactoryBot, DatabaseCleaner, RuboCop, Webpack**

- Led multi-version Rails and PostgreSQL upgrades across legacy enterprise systems.
- Refactored monolithic modules into service-oriented layers using PORO service objects and repository patterns.
- Migrated PostgreSQL schemas using zero-downtime migrations, backward-compatible constraints, and phased rollouts.
- Built automated regression coverage using RSpec, FactoryBot, DatabaseCleaner, and contract tests.
- Improved query performance using EXPLAIN ANALYZE, indexing strategies, and query refactoring.

TRUEPROJECTINSIGHT.COM | Feb 2020 - Mar 2023 - *Powered Project Governance Platform*

Role: **Technical Lead** | Stack: **Ruby on Rails, PostgreSQL, Angular, Sidekiq Pro, ActiveMQ, ElasticSearch, MicroStrategy, JSReport, Jenkins, Prometheus**

- Designed domain-driven core models for large-scale project governance datasets.
- Implemented multi-queue Sidekiq architecture (priority queues, rate limiting, scheduled jobs, dead-letter queues).
- Built messaging pipelines using ActiveMQ, with message transformation and idempotent consumers.
- Integrated ElasticSearch for advanced search and analytics indexing.
- Built enterprise reporting pipelines using MicroStrategy + JSReport with scheduled and parameterized reports.
- Implemented CI/CD pipelines using Jenkins, with parallel test execution and artifact versioning.
- Monitoring via Prometheus metrics + custom job telemetry.

MAHINDRA RETAIL | April 2018 - Jan 2020 - *High-Volume Campaign & Messaging Platform*

Role: **Technical Lead** | Stack: **Ruby on Rails, Angular, MySQL, RabbitMQ, Sidekiq, Twilio API, AWS EC2, Redis, REST APIs**

- Designed high-throughput coupon lifecycle engine with state-machine-driven workflows.
- Implemented async messaging using RabbitMQ + Sidekiq, decoupling campaign execution from delivery pipelines.
- Integrated Twilio SMS APIs with retry-safe delivery and rate-limiting logic.
- Built analytics pipelines aggregating campaign metrics using Redis-backed counters and batch jobs.

COMMONLIT.ORG | April 2018 - Jan 2020 — *Large-Scale EdTech Platform*

Role: **Senior Software Engineer** | Stack: **Ruby on Rails, PostgreSQL, React, Devise, Pundit, Stripe API, Heroku, CircleCI**

- Implemented authentication and authorization using Devise + Pundit RBAC.
- Built payment workflows using Stripe APIs with webhook-based event handling.
- Implemented background processing using multi-threaded job execution.
- Automated deployments using CircleCI, with blue-green deployment patterns on Heroku.

RIGHTBOAT.COM | Nov 2016 - Jan 2018 — *Distributed Marketplace Platform*

Role: **Senior Software Developer** | Stack: **Ruby on Rails, MySQL, ActiveMQ, Sidekiq, Nokogiri, AWS, Capistrano**

- Built marketplace ingestion pipelines using Nokogiri-based crawlers for external data sources.
- Implemented async workflows using Sidekiq + ActiveMQ for classification and enrichment pipelines.
- Automated deployments using Capistrano with AWS-based infrastructure.

EDUCATION

B-Tech. Electrical and Electronics Engineering.

Bharath University, Chennai

2005-2009



24 May 1986



Tamil & English



Chennai, India

DECLARATION

I hereby declare that the information furnished above is true.

By,

Arunkumar Kandasamy